Biochemical & Pharmacogenomic Divergence in MBD4-LOF Base Excision Repair Deficiency

UPSTREAM GLYCOSYLASE LOSS: Biallelic MBD4 Inactivation (Truncating LOF, NMD p_2s = 0.0367)

Failure to excise thymine from T:G mismatches at deaminated 5mCpG dinucleotides
and failure to excise misincorporated 5-substituted pyrimidine analogs (Gemcitabine / 5-FU)

AXIS 1: Pyrimidine Clearance Defect

- Intact MBD4 excises dFdC / 5-FU
- MBD4-LOF leaves analogs & T:G mismatches embedded in template
- Polymerase δ/ε stalling at fork

AXIS 2: Replication-Fork Stress

- Persistent T:G & dFdC lesions uncouple helicase from replicative polymerase
- Accumulation of RPA-coated ssDNA
- TP53 loss removes G1/S arrest

AXIS 3: No Nicked SSB Substrate

- Unlike XRCC1/Polβ loss (downstream), upstream MBD4 loss prevents APE1 backbone incision (no 5'-dRP nick)
- Zero PARP1 upregulation (TOST p = 0.020)

ACTIONABLE: Gemcitabine (dFdC)

- SL-XGBRanker Zero-Shot Rank #1
- Loads replication-fork lesions
- Sequential staggered dosing with ATRi (ATRiUM / TRESR precedent)

SYNTHETIC LETHAL: Ceralasertib (ATRi)

- TP53-Mutant: ΔLN_IC50 = -1.07 (d = -0.74) p_2s = 0.0060 (BH-FDR q = 0.036)
- Unstratified: ΔLN_IC50 = -0.73, p_2s = 0.0430
- MSS-Only: ΔLN_IC50 = -0.91, p_2s = 0.0307

CONFIRMED NULL: Single-Agent PARPi

- No PARP1-DNA trapping substrate
- Direct 5-Drug GDSC2 Evaluation (n=8 vs 478): Talazoparib (p=0.97), Olaparib (p=0.52), Composite PARPi Z-score p_2s = 0.841